

Unit 7 Seminar

Title: Data Science, Storage Solutions, and Big Data Processing

Icebreaker Activity (10 minutes) Where Do You See Data Science?

Example: Netflix uses data science to increase engagement and retention by tailoring recommendations to users. The ML algorithm predicts user behaviour based on history as well as similar users.

This includes, but is not limited to: the images shown to users and the titles, language and styles and genres recommended, additionally the speed of the connection, to dynamically change the quality of the stream, for lag-less streaming.

It can also use this data to predict what to create next to fulfill audience wants and how successful a series or film may be before they begin the creation.

It uses NoSQL to improve the speed of data collection and analytics. This also allows flexibility. E.g. different departments to access different types of information from the same source.

User data could be leaked. For example, a gay user may enjoy LGBTQ+ friendly TV and film, but may not wish for family or friends to know about their sexual identity.

2. Introduction to Data Science (15 minutes)

Statistics: Hypothesis testing, uncertainty,

Machine Learning: Prediction, Classification, Clustering

Domain Expertise: Industry Insights, User Expectation, Standards, Legal Implications, Competitive Analysis

Data-driven decision making - data > Guessing
Predictive analytics > use the past to predict the future
Machine Learning > pattern learning

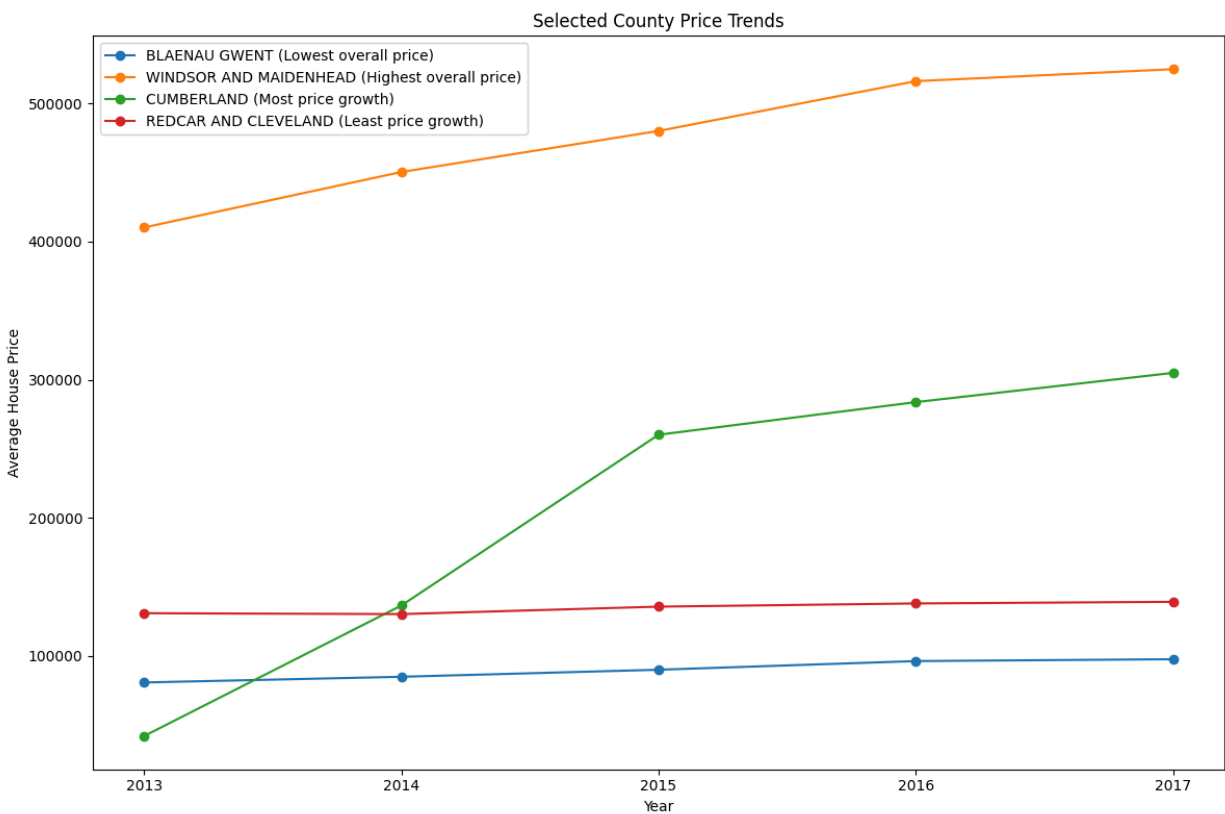
Forecast future demand, risk, services

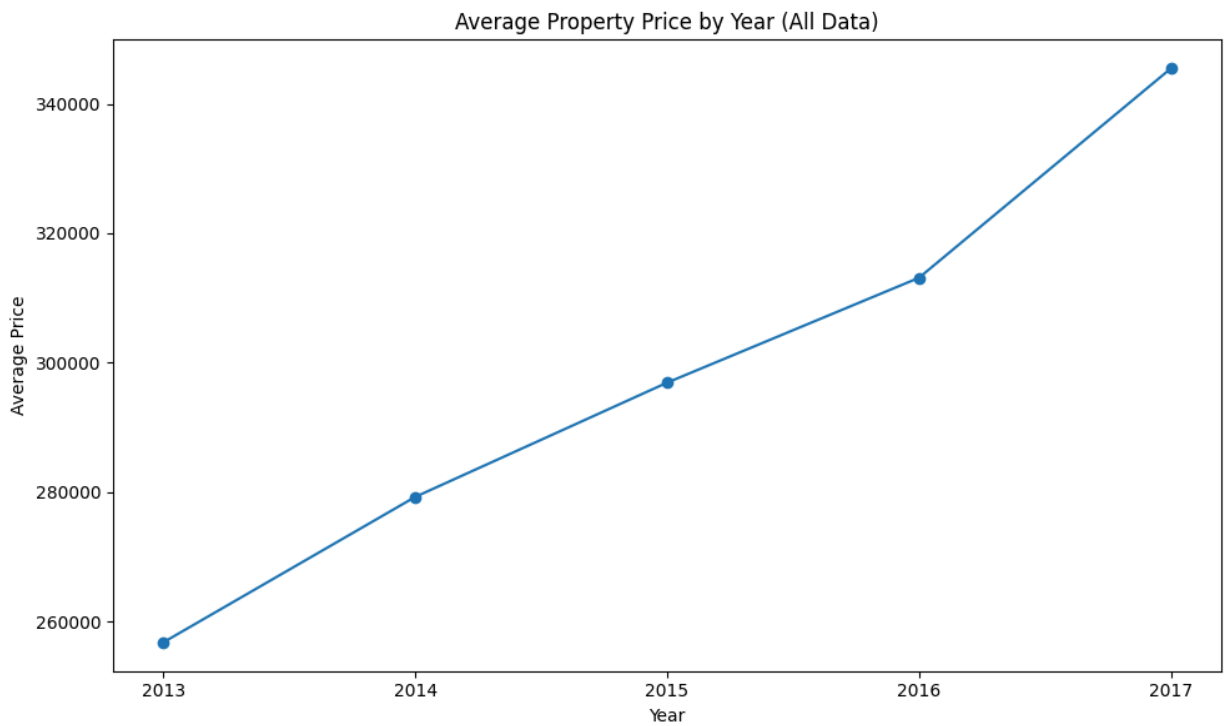
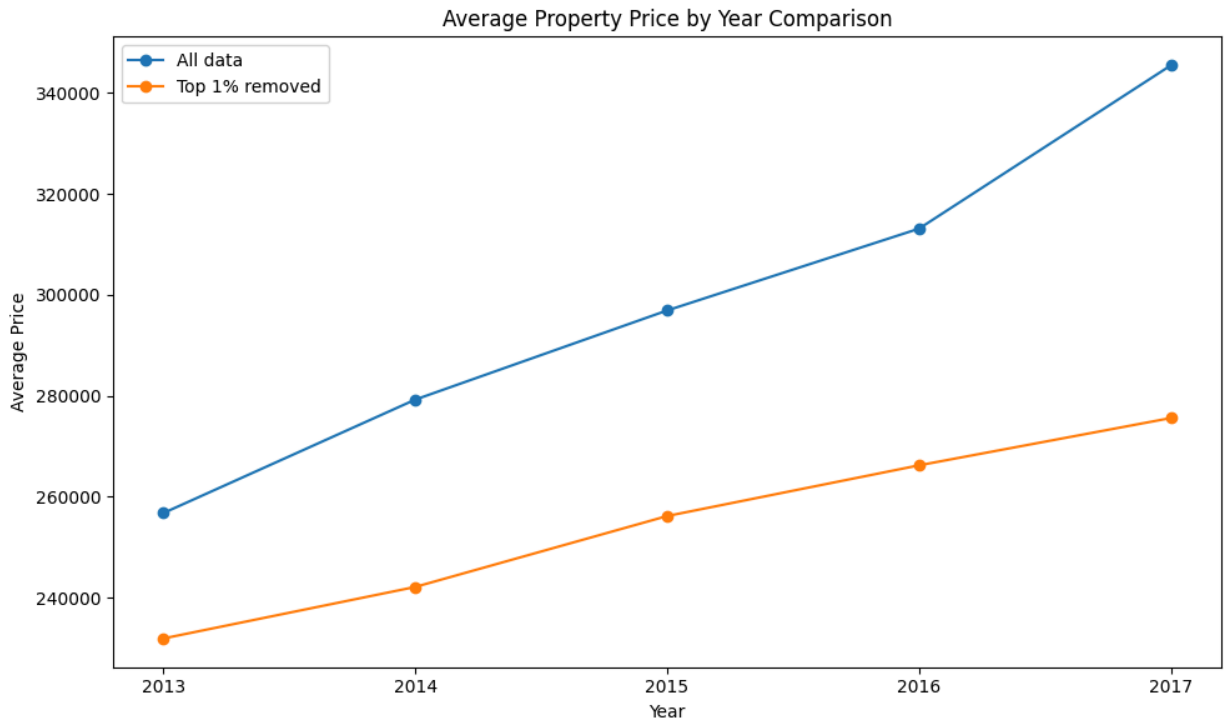
Legality, human considerations,

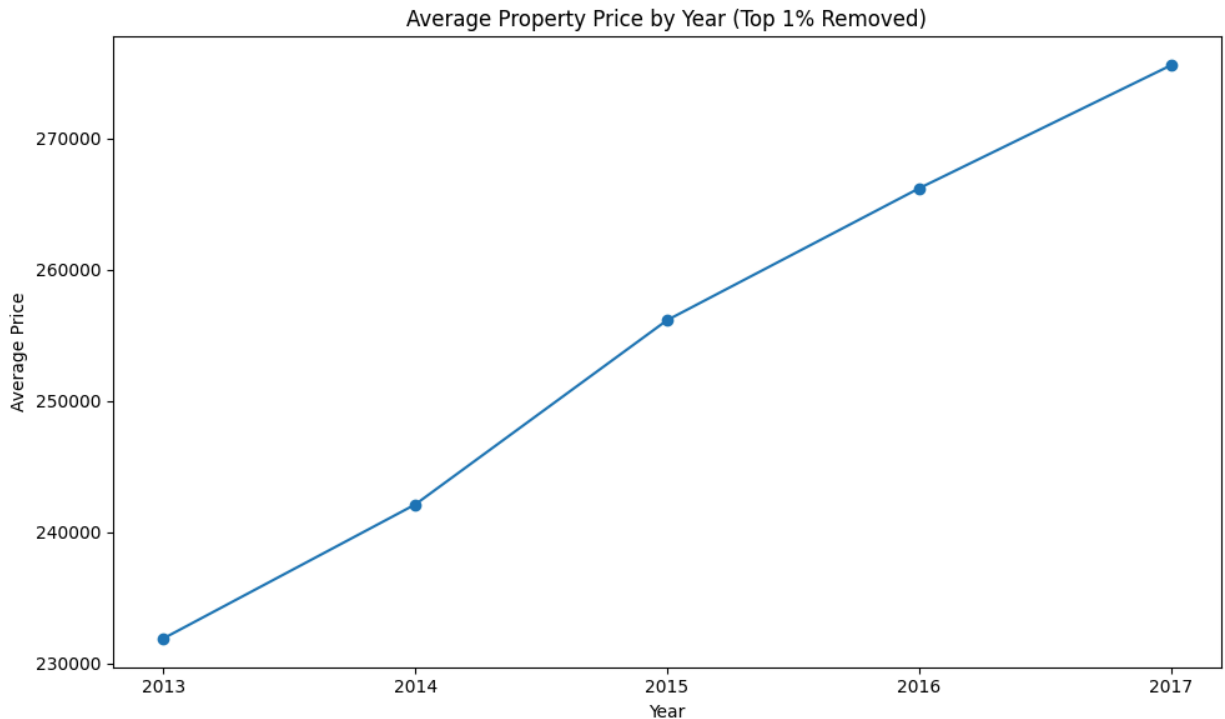
1. Collect - APIs, Apps, Sensors, Logs
2. Clean - Missing values, duplicated, formatting issues
3. Analyse - Patterns, models, segmentation
4. Visualise - Dashboard, charts, KPIs
5. Decide - Pricing, hiring, inventory, strategy, etc

Analyse a Simple Dataset

<https://github.com/feltpfaerie/Launch-Into-Computing/tree/master/coursework/house-price-data>







There was an unexpected duplication of data. The first cleaning algorithm produced unexpected results and the analysis suggested some properties were sold more than 600 times in four years. When I verified this, it was clear that the full address was not being considered and properties, like blocks of flats, were being identified as the same property. **This also highlighted the need for human verification and quality control of data generated from AI.**

There was also a stark difference in results when outliers (the top 1% were excluded from the data).

It's clear there is an upward trend, however this is more pronounced in the top 1% than other areas.

It's also possible to highlight the areas that have shown the most growth and the least. This may help identify where to invest next. Would require domain expertise.

Some properties were sold multiple time, (up to 7). There is either something wrong with the property, or something suspicious going on. This would require investigation and domain expertise.

3. Data Storage Solutions: SQL vs. NoSQL vs. Cloud Storage (20 minutes)

Students will explore different data storage models and their applications in modern computing.

Relational Databases (SQL):

Structured data is stored in tables with predefined schemas.

Used in banking, enterprise applications, and inventory management.

MySQL, PostgreSQL, Microsoft SQL Server.

NoSQL Databases:

Designed for scalability and flexibility, handling unstructured data.

Used in social media, IoT, and real-time analytics.

Example: MongoDB, Cassandra, Firebase.

Cloud Storage and Distributed Databases:

Data is stored across multiple data centres for scalability and fault tolerance.

Used in Big Data applications, AI training models, and enterprise systems.

Example: Google BigQuery, AWS S3, Apache HBase.

Activity: Students will compare SQL and NoSQL databases for a given scenario (e.g., an e-commerce platform handling structured vs. unstructured data).

4. Big Data Processing and Analytics (20 minutes)

Students will learn about Big Data processing frameworks and how businesses handle massive datasets efficiently.

Hadoop & MapReduce

A distributed computing framework that processes large datasets by breaking them into smaller chunks.

Used in financial fraud detection, genomic research, and weather forecasting.

Apache Spark

A real-time Big Data analytics engine that processes large-scale data faster than Hadoop.

Used in predictive maintenance, AI model training, and social media analytics.

Edge Computing & IoT Data Processing

Processing data closer to the source to reduce latency and bandwidth costs.

Used in self-driving cars, industrial automation, and smart city sensors.

Which data belongs in SQL and why?

banking transactions

birth, death, marriage register

Car registry - license plate, owner, MOT, etc

Census data

Data that must be correct

Data with the same structure

Which data belongs in NoSQL and why?

User behaviour

Social media

User reviews

Search data

Unstructured data in real time

Quick data retrieval

Inconsistent data from different pipelines

Large data sets - volume and velocity

Scalability

Why use a hybrid approach?

Combination of data types, some structured some not. E.g. user account, name, credit card, payment, etc.

Then combined with user behaviour for ML exercises.

Activity: Students will design a Big Data architecture for a company analysing social media trends in real time.

Allow brands to get a notification if their brand is trending, by region and then push an alert with a positive, negative or neutral sentiment. This will allow brands to respond quickly.

This service will scan multiple social media platforms:

X, Facebook, Instagram, TikTok, YouTube and combine the data

Some platforms provide APIs, some will need approved connectors or external data providers

Apache Kafka for high-throughput event streaming

Amazon S3 - “data lake”

NoSQL (Cassandra) - data coming in quickly, vast quantities and is unstructured.

Spark Streaming - integrates well with big data ecosystems, supports scalable stream processing, can combine streaming with ML pipelines

Checks sentiment, quantity, change - upward/downward trend/unusual pattern

Alerts - push notification and email

